# Aye Myat Myat Paing

## paper

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***Abstract*— The increasing number of vehicles and the limited availability of parking spaces in the city center create significant challenges for efficient parking management, especially in environments that require fast and reliable parking with limited computing resources. To address these challenges, a Smart Parking Management System (SPMS) is proposed that performs local parking-space occupancy detection using adaptive inverse image thresholding on a fog node located near the parking environment. It is enabling enhanced pixel density analysis within predefined parking regions under varying lighting conditions. To enhance resource efficiency, a fog computing architecture is integrated into the SPMS, where a fog node processes parking video streams close to the data source. Evaluating the effectiveness of the proposed approach, conventional global thresholding and adaptive inverse thresholding are compared using accuracy, precision, recall, and F1-score. Traditional and AI-based detection modes are also combined to support different parking-space detection approaches. The proposed fog-based adaptive-thresholding approach aims to provide an effective and resource-efficient solution for real-time smart parking management under various parking occupancy and lighting conditions.***

***Keywords*— Adaptive Inverse Threshold, Global Threshold, Image Processing, Computer Vision, AI-based Parking Space Detection, Fog Computing**

## I. Introduction

The increasing number of vehicles in downtown areas has faced many problems in searching and managing parking spaces because there is no real time information about available spaces and inefficient parking spaces. Drivers spend more time searching for available space, which can cause more loss of fuel and traffic congestions. Many organizations rely on manual parking management system using manpower. To monitor parking spaces and provide available information about parking status, the parking management system is proposed.

Camera ??? IP camera, -

Computer vision can be used to detect a parking space is occupied or empty from a camera. In this system, an image threshold is used to analyze the pixels inside predefined parking regions. A global threshold can be used for simple parking detection. But changes in lighting conditions are affected. To improve detection under different conditions, an adaptive inverse threshold is better. The pixel density inside each parking region is measured and determines the occupancy of parking spaces.

In addition, fog computing is used in the proposed SPMS to process video data closer to the parking environment. And the fog node performs the parking-space detection locally. It can help reduce processing delay and unnecessary data transmission. And then it can also support real-time parking monitoring when resources are limited.

The proposed system has a real-time dashboard that shows parking usage statistics such as total parking slots, available slots, occupied slots, and occupancy percentage for effective parking management rather than the traditional way. SPMS includes both traditional and AI-based detection modes to support different parking detection approaches. The traditional mode uses image processing and threshold-based pixel density analysis.

Another approach for parking space detection is AI-based. Those approaches to adaptive inverse thresholding are evaluated using accuracy, precision, recall, and F1. It provides reliable parking occupancy detection under bright, low-light, and shadow conditions.

The structure of the paper comprises six sections: Section II sets a literature review on relevant research studies. In Section III, implementation methodology details are explained. Next, Section IV outlines the implementation/experiments, while Section V presents evaluation and Section VI concludes.

## II. Related Works

The demand for parking in public locations is overwhelming, causing traffic jams inside the parking lots. To solve this problem, numerous studies have implemented. P. Rajesh et al. [1] proposed a novel intelligent parking system that utilizes edge computing and computer vision with a single monocular camera infrastructure. This system guaranteed real-time operation, essential for efficient navigation and space management, by reducing latency through on-site data processing using edge computing.

Therefore, this study proposes xxxx

## III. Methodology

The proposed xxx introduces

Fig. 1. Proposed Fault-Tolerant system architecture

Figure 1 illustrates

### *A. Dask-Based Fault-Tolerant* Distributed *Architecture*

The proposed system uses Dask's cluster management to

.

### *B. Adaptive Timeout-Based Worker Failure Detection*

The system proposes an adaptive mechanism for failure

## IV. Experimental Results

### *A. Experimental Setup*

XXXXX

TABLE I. WORKER RELIABILITY AND RECOVERY SELECTION FOR 5W–5F

| *Worker* | *Success* | *Failure* | *Score* | *CPU(%)* |
|---|---|---|---|---|
| Worker-1 | 3 | 0 | 3 | 37.5 |
| Worker-2 | 1 | 0 | 1 | 2.0 |
| Worker-3 | 1 | 0 | 1 | 25.0 |
| Worker-4 | 0 | 1 | 0 | 0.0 |
| Worker-5 | 0 | 1 | 0 | 0.0 |

Table III shows

## V. Comparison with the Standard Dask Baseline

XXXX

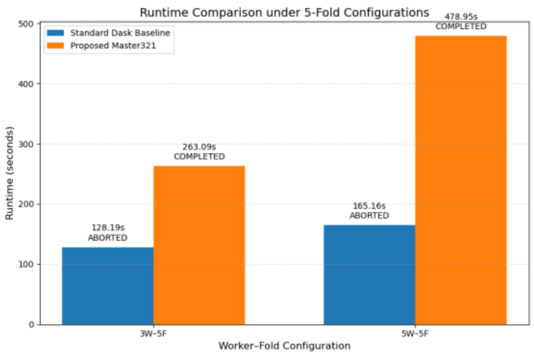


Fig. 2. Runtime comparison under 5-Fold configurations

Figure 2 shows the comparison of fault tolerance performances of the Dask Baseline and the proposed development in the 3W–5F and 5W–5F configurations. The Dask Baseline was aborted after Fold 1 and Fold 2 passed the predetermined timeout period in the two configurations. However, the framework identified the tasks that were delayed using the adaptive timeout, chose appropriate recovery workers based on reliability and CPU load, and after making these selections, reallocated the tasks. This way, it managed to recover all the failed tasks and finish the five folds successfully without stopping the work.

## VI. Conclusion

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